

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer 2025

Course: B.Tech.

Branch : Civil Engineering

Semester : Semester VII

Subject Code & Name: BTCVE705A_Y20 & Construction Techniques

Max Marks: 60

Date: 25-07-2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level /CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	The basic parts of clam shell bucket are a. Closing line b. Hoist line c. Sheaves, bracket d. All of these	CO1	
2	...refers to the collection of innovative tools, machinery, modifications, software, etc a. Construction technology b. Construction techniques c. Construction management d. None of these	CO1	1
3	Operations such as excavation, digging of large quantities of earth, moving, compacting, levelling etc. are performed using a. Transportation equipment b. Earth moving equipment c. Excavation equipment d. Earth filling equipments	CO1	1
4	BULLDOZERS is also known as a. Crawler tractors. b. Backhoe c. Power shovels. d. None of these	CO2	1
5	A backhoe , also called a a. Rear actor b. Back actor c. Rear actor or back actor d. Only backhoe	CO2	1
6	Following are the Methods of tunneling in rock a. Full face method b. Heading and bench method c. Drift method d. All of these	CO2	1
7	Following are the Health risk Jack hammer a. Tendonitis-EFFECT ON BONE b. Hearing loss c. Lower back pain d. All of these	CO3	1
8	The Prefabrication certain stages of preparation a. Casting b. Curing c. Transportation and erection d. All of these	CO3	1

9	Prefabrication is the				CO4	1
	a. Dismantal components	b. Weldind components	c. practice of assembling components	d. Joint components		
10	Prefabricated components is				CO4	1
	a. Columns	b. Beam	c. All of these	d. None Of these		
11	Drum mix plant components				CO5	1
	a. Control van	b. Binder storage tank	c. Drum mixer	d. All of these		
12	Construction accidents are				CO5	1
	a. Construction Site Falls	b. Crane Accidents	c. Electrical Accidents	d. All of these		
Q. 2	Solve the following.					12
A)	Enlist the various types of Rollers and explain any one.				CO1	6
B)	What is bulldozer? State the operation carried out by bulldozer and factors affecting the selection of bulldozer				CO1	6
Q.3	Solve the following.					12
A)	Discuss the role of compressors in excavation operations?				CO2	6
B)	Define jack hammers. Explain the working of jack hammer				CO2	6
Q. 4	Solve Any Two of the following.					12
A)	What are the methods and equipment used for underwater concreting?				CO3	6
B)	Explain the operations involved in RMC plant				CO3	6
C)	Define grouting. Describe the various materials used for grouting. Write down advantages of grouting				CO3	6
Q.5	Solve Any Two of the following.					12
A)	Write down advantages and disadvantages of pre-fabrication				CO4	6
B)	Discuss the advantages and limitations of using different types of cranes for heavy lifting tasks				CO4	6
C)	What do you mean by Hoisting Equipment? Explain Rail mounted tower crane with suitable diagram.				CO4	6
Q. 6	Solve Any Two of the following.					12
A)	What safety measures should be implemented to prevent accidents in construction projects?				CO5	6
B)	What are the causes of accidents on construction site? How to prevent it.				CO5	6
C)	Explain construction method of diaphragm walls.				CO5	6